

smartphones will be sold by 2024 end. While this technology may not match terrestrial cellular services in speed and latency, it offers a unique solution for connectivity in currently underserved areas, potentially bridging the coverage gap in remote regions and creating new opportunities for rural businesses and individuals.

The evolution of satellite connectivity in smartphones

Before Apple introduced satellite connectivity, T-Mobile had already partnered with SpaceX to offer similar services through Starlink satellites. This collaboration eliminates the need for dedicated satellite phones by using T-Mobile's 1900MHz spectrum, allowing existing 5G smartphones to connect for SMS, MMS, and selected messaging apps, with plans to include voice and data services in the future.

In January 2023, Qualcomm launched Snapdragon Satellite for Android, which provides two-way messaging like Apple's emergency SOS feature. Initially available on the Snapdragon 8 Gen 2 chipset, this technology requires specific hardware, limiting its initial availability to new devices. Qualcomm's service, in collaboration with Iridium, currently covers North America and Europe.

MediaTek has introduced satellite connectivity chipsets with two SoCs designed for two-way satellite communication on smartphones. These chipsets can be integrated into any 4G or 5G smartphone, regardless of the primary SoC, making satellite connectivity more accessible.

Samsung is also entering the satellite connectivity arena with its Exynos modems. The Exynos 5300 modem is expected to enable two-way satellite messaging and may feature in the upcoming Galaxy S24 series smartphones.

Do we really need satellite communication in our smartphones?

In remote locations lacking terrestrial cell towers, satellite communication offers a solution by connecting phones directly to satellites. This technology addresses the challenge of insufficient physical infrastructure, potentially achieving near-universal coverage. While satellite phones have been essential for explorers and military personnel, current smartphone capabilities are limited to short messages, which may not fully meet the needs of regular mobile internet users. Although satellite internet is a valuable complement to traditional network services, it is not yet a replacement for existing network coverage models.

In India, the market for satellite technology is significant due to challenging terrain and developing infrastructure. Reliance Jio's JioSpaceFiber, India's first satellite broadband service, offers fibre-optic-like speeds and reliable high-speed internet nationwide. Similarly, Airtel's joint venture with OneWeb, Airtel OneWeb, has received regulatory approval to provide satellite services, promising nationwide coverage in the near future.

Advancements and regulations in India's mobile-to-satellite connectivity

The Department of Telecommunications (DoT) has decided against allowing satellite Earth station gateways in urban areas due to potential interference with mobile networks. This decision contrasts with the Telecom Regulatory Authority of India's (TRAI) recommendation to set up satellite gateways anywhere within India.

The recent passage of the Telecom Bill, 2023, introduces structural changes to the telecom regulatory framework, including simplifying licensing, clarifying spectrum assignment, and implementing stringent user verification requirements. Key measures include allocating satellite broadband spectrum, mandating biometric authentication for new telecom users, regulating encryption standards, and granting the government power to control telecom networks during war time.

The decision to allocate satellite broadband spectrum rather than auctioning it is expected to expedite service launches by major players like Bharti Group-backed OneWeb, Reliance's Jio Satellite Communications, Elon Musk's Starlink, and Amazon's Project Kuiper. These companies will be granted a global mobile personal communication by satellite (GMPCS) license.

The satellite communication (satcom) market in India is emerging, with significant potential, especially in rural and remote areas. Companies like Reliance Jio, Starlink, Amazon, Bharti-backed OneWeb, and Tata-Telesat are working on satellite constellations to deliver broadband-from-space services. Recent discussions on the mode of allotment of satellite spectrum have generated significant industry debate.

The satellite internet race

As companies strive for rapid satellite connectivity solutions, leveraging existing infrastructure becomes crucial. Startups like Skylo and ESat Global are leading this trend, while phone manufacturers and network operators explore partnerships. Traditional GEO operators such as Intelsat, Viasat, and SES face challenges from SpaceX and OneWeb as the shift to mobile phone connectivity opens new markets.

Recent developments underscore the intense competition in satellite connectivity. Companies must inno-